Campus Connection

Implementation summary for presentation

> **In brief:** Added sample users with roles, student and course seed data, sample course registrations, and repeat-safe seeding. A test confirms running the seeder twice does not increase the seeded record counts.

What was requested and implemented

Requirement	What the project does now
Identify what was added / which question	The Laravel `User` model by itself was standard authentication boilerplate; it did not identify a coursework question. The later data-seeding requirements were implemented explicitly.
Seed at least five students and four courses	`DatabaseSeeder` seeds five student records and four course records with sample academic details.
Create sample user accounts and assign roles	Seeds three demo accounts assigned the `admin`, `student`, and `lecturer` role values. The user table has a `role` column, and the User model allows that attribute.
Seed course registrations	Creates one registration for each of the five seeded students, pairing them across the four seeded courses.
Avoid unnecessary duplicate records on rerun	Uses `firstOrCreate` keyed by unique user email, student number, course code, or student/course pair. Existing seeded rows are reused on subsequent runs.

Seeded data at a glance

3	5	4	5
sample users with roles	students	courses	course registrations

Each student receives at least one sample course registration. The seed data covers the minimum requested quantities.

Files changed

- `database/seeders/DatabaseSeeder.php` - Creates the sample accounts, students, courses, and registrations using repeat-safe lookups.
- `database/migrations/2026_11_04_010000_add_role_to_users_table.php` - Adds a role field to users, defaulting to `student`.
- `app/Models/User.php` - Adds `role` to the model's mass-assignable attributes.
- `tests/Feature/DatabaseSeederTest.php` - Runs the seeder twice and checks the expected row counts remain 3 users, 5 students, 4 courses, and 5 registrations.

Verification

> **4 tests passed, 10 assertions.**
> This included the idempotency check and all three existing course-registration feature tests.

Test command used: `vendor\bin\pest.bat tests\Feature\DatabaseSeederTest.php tests\Feature\CourseRegistrationTest.php`

How to present the result

1. Explain the sample data and role field.
2. Show the seeder's `firstOrCreate` keys and explain that they prevent adding the same seeded records on every run.
3. Show the test that runs `DatabaseSeeder` twice and verifies counts do not grow.

Important scope note

The role values are stored on sample user accounts; this change does not add route protection, permission checks, or a full authorization package. The seeded accounts and passwords are for local demonstration only and must be changed before any real deployment.

To apply the role migration and seed the database: `php artisan migrate --seed`

Prepared from the implemented project files and test results. This summary describes the work completed; it is not a claim that the coursework PDF was fully reviewed.